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Studierichting: Informatica

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$$\int \frac{x^3 + 2x - 1}{x^3 - x^2 + x - 1} dx$$

$$x^3 - x^2 + x - 1 = (x - 1)(x^2 + 1)$$

$$\frac{x^3 + 2x - 1}{(x - 1)(x^2 + 1)} = 1 + \frac{x^2 + x}{(x - 1)(x^2 + 1)} = 1 + \frac{A}{x - 1} + \frac{Bx + C}{x^2 + 1}$$

$$A = \left. \frac{x^2 + x}{x^2 + 1} \right|_{x=1} = 1$$

$$\left. \frac{x^2 + x}{x - 1} \right|_{x=i}$$

$$= \frac{i - 1}{i - 1}$$

$$= 1 = C \Rightarrow B = 0$$

$$\Rightarrow \int \left(1 + \frac{1}{x - 1} + \frac{1}{x^2 + 1} \right) dx$$

$$= x + \ln |x - 1| + \text{Bgtan } x + k$$

References